

Tutorial 5: Belts

In-class Activity

1. What are the common types of belts used in mechanical systems?
2. What are the advantages and disadvantages of using belts for power transmission?
3. How do you determine the appropriate belt size for a specific application?

Calculations

Question 1

A plastic flat belt 60 mm wide and 0.5 mm thick transmits 10 kW. The input pulley has a diameter of 300 mm, it rotates at 2800 rpm, and the output pulley speed is 1600 rpm; the pulleys are 700 mm apart, the coefficient of friction is 0.2, and belt weight is 25 kN/m³. It is assumed that the driver is a high torque motor, and the driven machine is under a medium shock load. Determine;

- (i) the torque at smaller pulley
- (ii) the contact angle at driver and driven pulleys
- (iii) the centrifugal tension in the belt
- (iv) the maximum tension and stress in the belt

Example Solution

Given Data:

Belt width, $b = 60 \text{ mm} = 0.06 \text{ m}$

Belt thickness, $t = 0.5 \text{ mm} = 0.0005 \text{ m}$

Power, $P = 10 \text{ kW} = 10000 \text{ W}$

Diameter of smaller pulley, $d_1 = 300 \text{ mm} = 0.3 \text{ m}$

Speed of smaller pulley, $n_1 = 2800 \text{ rpm}$

Speed of larger pulley, $n_2 = 1600 \text{ rpm}$

Distance between pulleys, $c = 700 \text{ mm} = 0.7 \text{ m}$

Coefficient of friction, $f = 0.2$

Belt weight, $w = 25 \text{ kN/m}$

i - Torque at smaller pulley

$$\begin{aligned} \text{Power, } P &= \frac{2\pi n_{rpm} T}{60} \\ T &= \frac{(10000)60}{2\pi(2800)} \\ &= 34.1 \text{ Nm} \end{aligned}$$

ii - Contact angle

$$\sin \alpha = \frac{r_2 - r_1}{c}$$

Diameter of larger pulley is unknown, thus we can find it using the relation between the speeds of the pulleys.

$$\begin{aligned}\frac{n_1}{n_2} &= \frac{d_2}{d_1} \\ d_2 &= \frac{n_1 d_1}{n_2} = \frac{2800 \times 0.3}{1600} = 0.525m\end{aligned}$$

where, smaller pulley radius, $r_1 = \frac{0.3}{2} = 0.15m$

larger pulley radius, $r_2 = \frac{0.525}{2} = 0.2625m$

Find α :

$$\begin{aligned}\sin \alpha &= \frac{0.2625 - 0.15}{0.7} \\ \alpha &= 9.25^\circ\end{aligned}$$

Equation to find contact angle is in radians $\phi = \pi - 2\alpha$, thus we need to convert to radians:

$$\alpha = 9.25^\circ \times \frac{\pi}{180} = 0.1614rad$$

Contact angle for driver pulley(smaller pulley):

$$\begin{aligned}\theta_1 &= \pi - 2\alpha \\ &= \pi - 2(0.1614) = 2.82rad\end{aligned}$$

Contact angle for driven pulley(larger pulley):

$$\begin{aligned}\theta_2 &= \pi + 2\alpha \\ &= \pi + 2(0.1614) = 3.48rad\end{aligned}$$

iii - Centrifugal tension in the belt

Centrifugal tension, $F_c = \frac{w}{g}v^2$, or $F_c = \rho A v^2$

where w is belt weight per unit length (N/m) and v is velocity of belt. g is acceleration due to gravity.

Belt weight given as $25kN/m^3 = 25000N/m^3$. Need to find belt weight per unit length, w in (N/m):

$$\begin{aligned}w &= 25000 \left(\frac{N}{m^3} \right) \times \text{Cross section area of belt}(m^2) \\ &= 25000 \left(\frac{N}{m^3} \right) \times 0.06m \times 0.0005m = 0.75N/m\end{aligned}$$

where,

Belt width, $b = 60 \text{ mm} = 0.06 \text{ m}$

Belt thickness, $t = 0.5 \text{ mm} = 0.0005 \text{ m}$

Next, velocity of belt, $v = \frac{\pi d_1 n_1}{60} = \frac{\pi \times 0.3 \times 2800}{60} = 43.98 \text{ m/s}$

Now, calculate centrifugal tension, F_c :

$$\begin{aligned} F_c &= \frac{0.75}{9.81} (43.98)^2 \\ &= 147.88 \text{ N} \end{aligned}$$

iv - Maximum tension and stress in the belt

Maximum tension is tension of the belt at tight side, F_1

Using belt tension equation for flat belt:

$$\frac{F_1 - F_c}{F_2 - F_c} = e^{f\phi}$$

where,

F_1 : tension on the tight side in the belt,

F_2 : the tension on the slack side of the belt,

F_c : centrifugal tension,

f : coefficient of friction,

ϕ : contact angle in radians.

$$\begin{aligned} \frac{F_1 - 147.88}{F_2 - 147.88} &= e^{0.2(2.8188)} \\ \frac{F_1 - 147.88}{F_2 - 147.88} &= 1.7573 \\ F_1 - 147.88 &= 1.7573(F_2 - 147.88) \\ F_1 &= 1.7573F_2 - 259.89 + 147.88 \end{aligned}$$

$$F_1 = 1.7573F_2 - 112.01 \quad (1)$$

From torque equation:

$$\begin{aligned} T &= (F_1 - F_2)r_1 \\ 34.1 &= (F_1 - F_2)(0.15) \\ \frac{34.1}{0.15} &= F_1 - F_2 \\ 227.33 &= F_1 - F_2 \end{aligned}$$

$$F_2 = F_1 - 227.33 \quad (2)$$

Substitute equation 2 into equation 1:

$$\begin{aligned}
 F_1 &= 1.7573(F_1 - 227.33) - 112.00 \\
 F_1 &= 1.7573F_1 - 399.5 - 112.00 \\
 F_1 - 1.7573F_1 &= -511.5 \\
 -0.7573F_1 &= -511.5 \\
 F_1 &= \frac{511.5}{0.7573} \\
 &= 675.42N
 \end{aligned}$$

OR use belt tension relationship:

$$F_1 = F_c + \left(\frac{\gamma}{\gamma - 1} \right) \frac{T_1}{r_1}$$

where,

$\gamma = e^{f\phi}$ for flat belt.

Substituting the values:

$$\begin{aligned}
 \gamma &= e^{0.2(2.8188)} = 1.7573 \\
 F_1 &= 147.88 + \left(\frac{1.7573}{1.7573 - 1} \right) \frac{34.1}{0.15} \\
 F_1 &= 675.48N
 \end{aligned}$$

The final answer is approximately the same using both methods.

Maximum stress in the belt, A is the cross-sectional area of the belt:

$$\begin{aligned}
 \sigma_{max} &= \frac{F_1}{A} \\
 &= \frac{675.48}{0.06 \times 0.0005} \\
 &= 22.52MPa
 \end{aligned}$$

Question 2

V belt drive has a 200mm diameter small sheave with 170° contact angle, 38° included angle, coefficient of friction of 0.15, 1600rpm driver speed, belt weight 8N/m and a tight side tension of 3kN. Determine the power capacity of the drive.

Question 3

A v belt drives has an included angle of 38° , belt density, $\rho = 2100 \text{ kg/m}^3$, belt cross sectional are of 145 mm^2 , coefficient of friction of 0.25, $r_1=150 \text{ mm}$, $\phi=160$, $n_1=3000 \text{ rpm}$ and $F_2=800 \text{ N}$. Calculate

- (i) the maximum power transmitted
- (ii) the maximum stress in the belt

Question 4

In a horizontal V belt drive for a centrifugal blower, the blower is belt driven at 600 rpm by a 15 kW, 1750rpm electric motor. The groove angle of the pulley is 35° . The center distance is twice the diameter of the larger pulley and velocity of the belt is 20 m/s. The density of the belt material is 1500 kg/m^3 and maximum allowable stress is 4MPa. The coefficient of friction of motor pulley is 0.5. Determine

- (i) the diameter of the blower pulley
- (ii) the belt length
- (iii) the size of the belt in mm²

Question 5

A motor with 16 kW power is attached to a compressor via a flat belt with a length of 3 m. The belt experienced a maximum stress of 4 MPa in an operation and medium shock with $K_s = 1.5$. If the coefficient of friction between the belt surface and the pulleys are 0.4 and the density of the belt is 950 kg/m³ and the details of the belt drive are tabulated in Table 1. Determine;

	Diameter	Speed
Compressor Pulley	0.6	1200
Motor Pulley	dm	1800

Table Q5

- (i) the diameter of the motor pulley, dm
- (ii) the center distance, c between the pulleys.
- (iii) angle of wrap or contact angle, of both pulleys. Does the angle equal? In what circumstances the angles will be equal?
- (iv) torque on both pulleys. Does the torque on both pulleys equal? Why? In what circumstances the torque will be equal? How about power?
- (v) Velocity of the belt, v
- (vi) Centrifugal force, F_c
- (vii) Cross section of belt, A
- (viii) Tight tension on belt, F_1
- (ix) Slack tension on belt, F_2
- (x) The initial tension on the belt, F_i

Well-defined problem

1. If the area of the belt is reduced to $A = 0.0001 \text{ m}^2$, suggest the number of belts required using the same tensions in Question 1.
2. Compare the power capacity of the belt drive if the flat belt is replaced by another belt as detailed below:

Belt Types	Specific Weight (kg/m^3)	σ (MPa)	CoF	Dimension	Included angle
V-belt	950	4	0.4	$A=2000\text{mm}^3$	38°
Round belt	950	4	0.4	d=2mm	-

Table Q2 Belt Types Comparison

3. Compare the value of the belt area in Question 1 if the speed of the motor is reduced to 20%. How about the belt numbers?

Answer

Q2

(i) $P = 34.6kW$

Q3

(i) $P = 44.0kW$

(ii) $\sigma_{max} = 11.96MPa$

Q4

(i) $d_2 = 0.636m = 636mm$

(ii) $L = 3.92m$

(iii) $A = 223mm^2$

Q5

(i) $d_1 = 0.4m = 400mm$

(ii) $c = 0.708m$

(iii) $\phi_1 = 163.76^\circ, \phi_2 = 196.24^\circ$

(iv) $T_1 = 84.88Nm, T_2 = 127.32Nm$

(v) $v = 37.70m/s$

(vi) $F_c = 1350225.5A, F_c = 638.66 \text{ N}$ (After find A)

(vii) $A = 473mm^2$

(viii) $F_1 = 1261.42N$

(ix) $F_2 = 837.19N$

(x) $F_i = 1049.31N$